

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		At least 4 field lines are radial and symmetrical (1) At least 2 equipotentials are circles / spheres (1) Out of positron and into anti-neutron (1)	3			3		
	(b)		Indicative content: Differences 1. electric fields - charges & grav fields – masses (or force per unit mass, force per unit charge or forces between charges & masses) 2. e-fields – into & out, g-fields only into (direction point – also accept grav forces only attractive, e-forces attractive & repulsive) 3. Gravity is far weaker (on small scale) OR electrical negligible on large scale (i.e. scale point) 4. E potential work done per unit charge & G work done per unit mass 5. E-fields depend on permittivity OR g-fields depend on dark matter Similarities 1. Same shape field (for point/spherical masses/charges). Note: this point is different from the same shape graph which is the next point. 2. inverse square law (applies to both) 3. Both fields lead to (concept of) PE 4. Potential or PE is inversely proportional to distance 5. Both fundamental force laws 6. Both have scalar fields OR both have vector fields (or both!) 7. Anything to do with exchange particles (photon, gluon, bosons) NOTE: some differences can be viewed as similarities and vice versa e.g. energy per unit charge and energy per unit mass could be considered a similarity. Also, the different exchange particles could be considered a difference.	6			6		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			5-6 marks Similarities and differences between electric and gravitational fields covered comprehensively. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Similarities or differences between electric and gravitational fields covered comprehensively or limited description of both. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks Limited description of either the similarities or the differences. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks No attempt made or no response worthy of credit.						
			Question 2 total	9	0	0	9	0	0